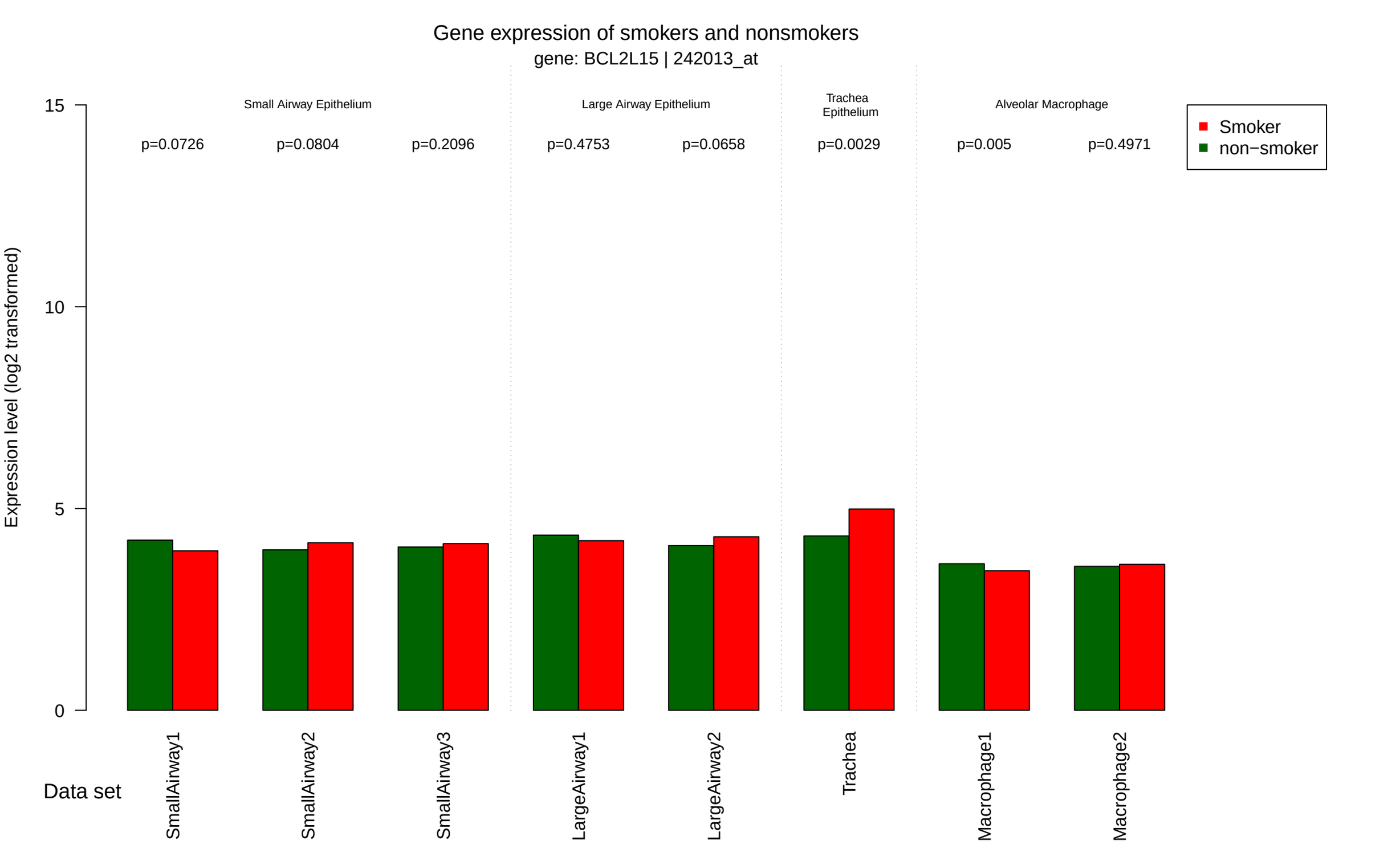
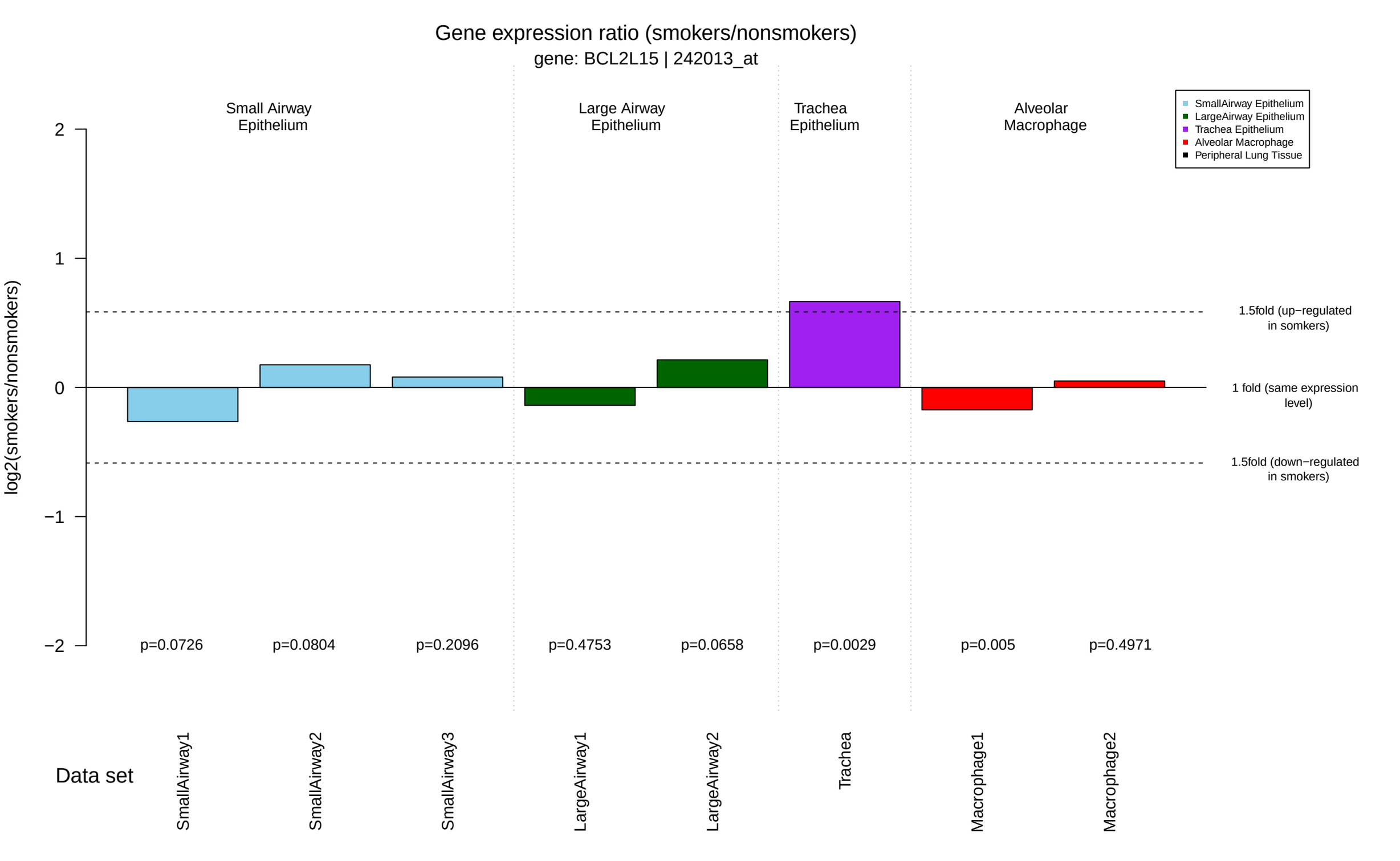
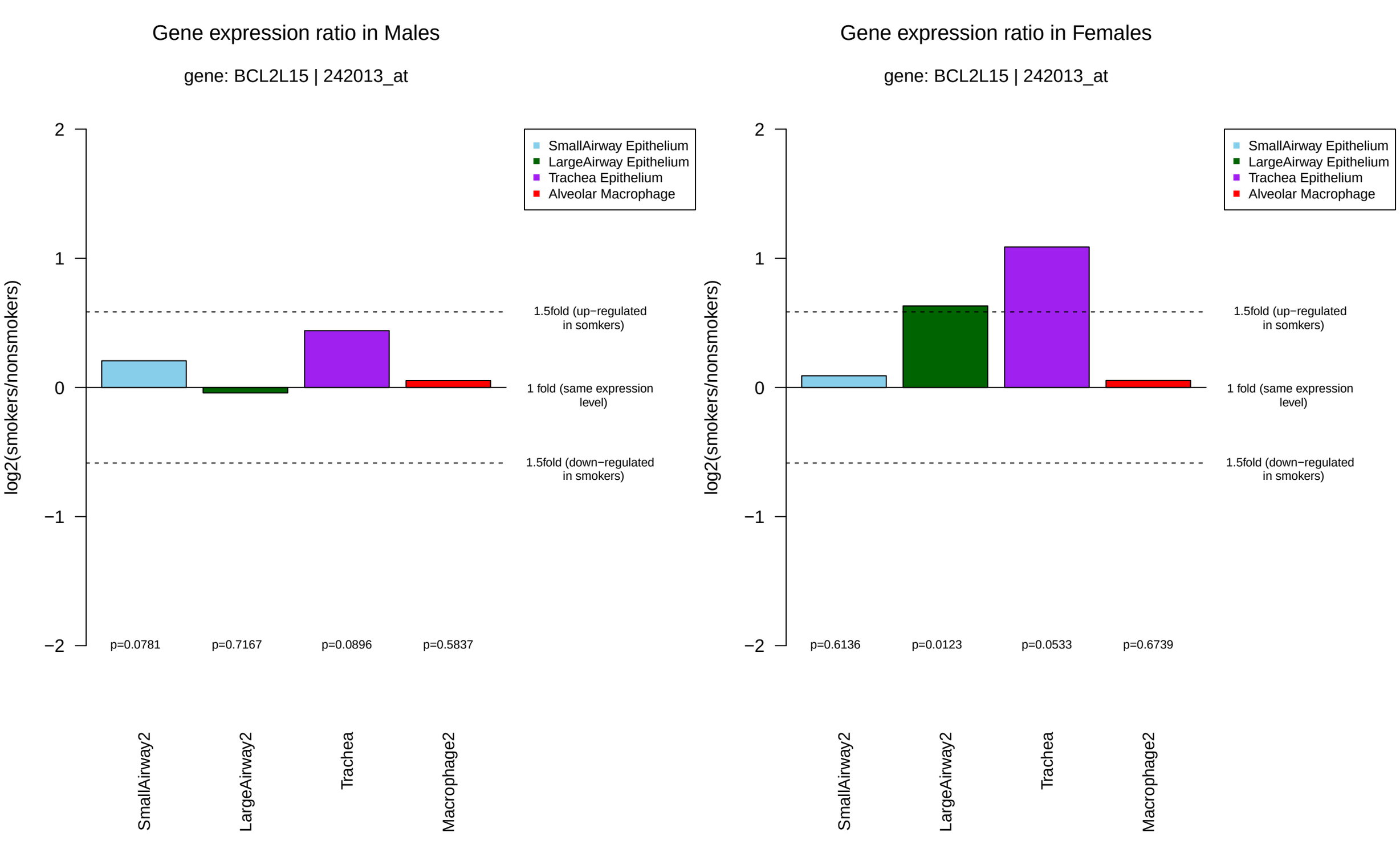
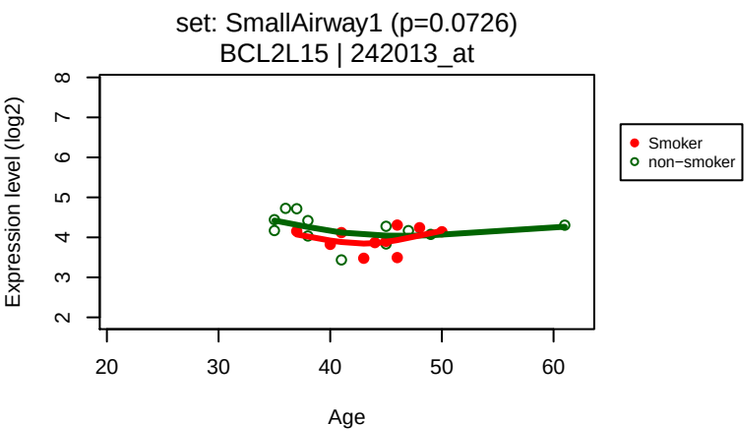
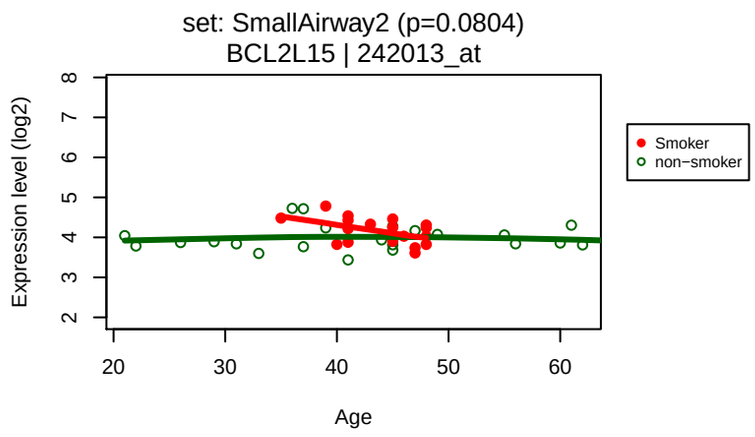
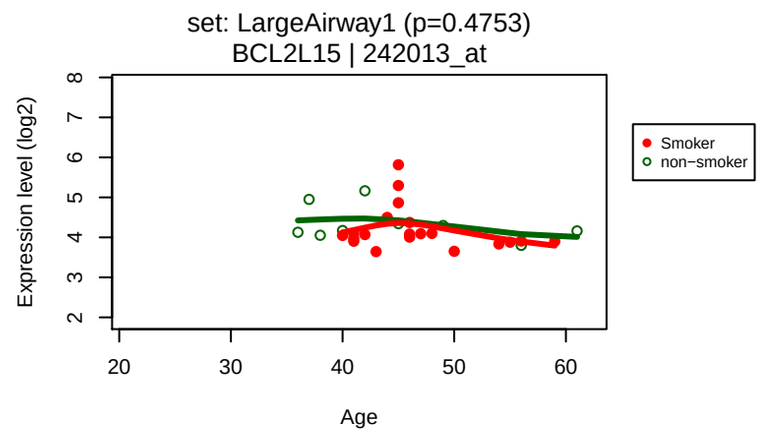

Gene expression of smokers and nonsmokers
gene: BCL2L15 | 242013_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0726
p=0.0804
p=0.2096
p=0.4753
p=0.0658
p=0.0029
p=0.005
p=0.4971
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

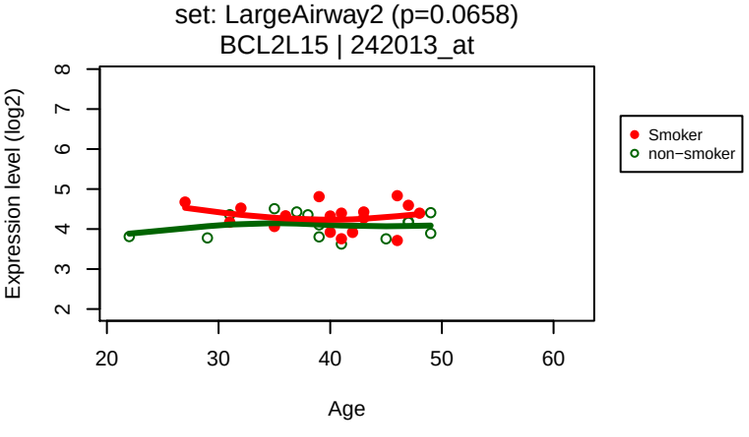

Gene expression ratio (smokers/nonsmokers)
gene: BCL2L15 | 242013_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0726
p=0.0804
p=0.2096
p=0.4753
p=0.0658
p=0.0029
p=0.005
p=0.4971
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

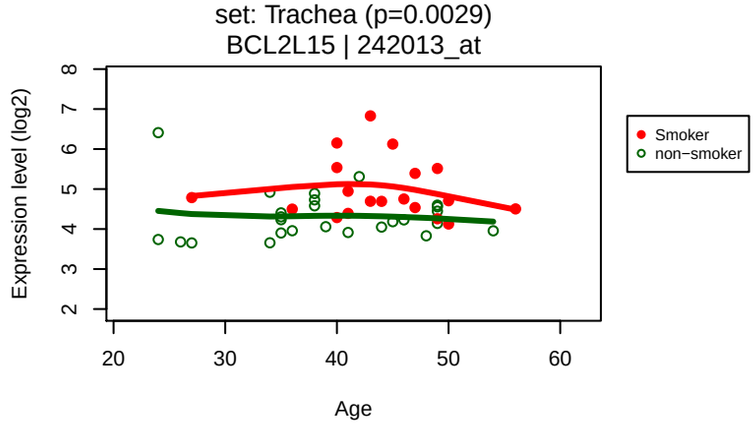

Gene expression ratio in Males
gene: BCL2L15 | 242013_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0781
p=0.7167
p=0.0896
p=0.5837
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BCL2L15 | 242013_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6136
p=0.0123
p=0.0533
p=0.6739
SmallAirway2
LargeAirway2
Trachea
Macrophage2

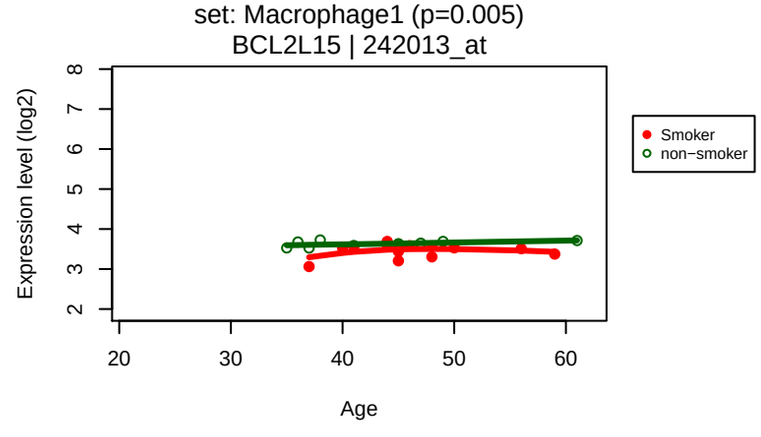

set: SmallAirway1 (p=0.0726)
BCL2L15 | 242013_at
Expression level (log2)
Age
Smoker
non−smoker

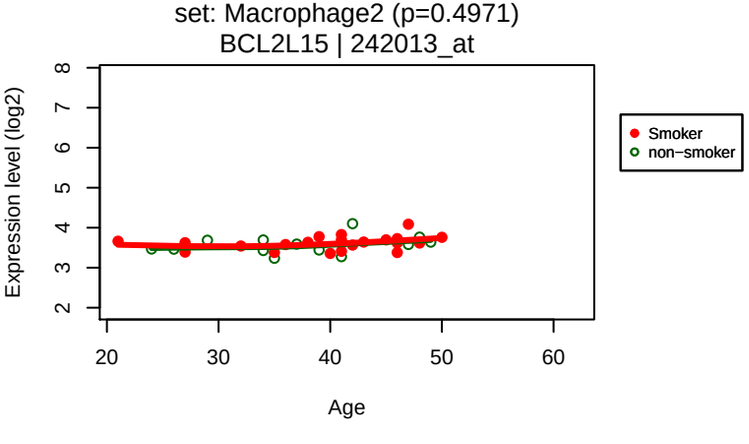

set: SmallAirway2 (p=0.0804)
BCL2L15 | 242013_at
Expression level (log2)
Age
Smoker
non−smoker

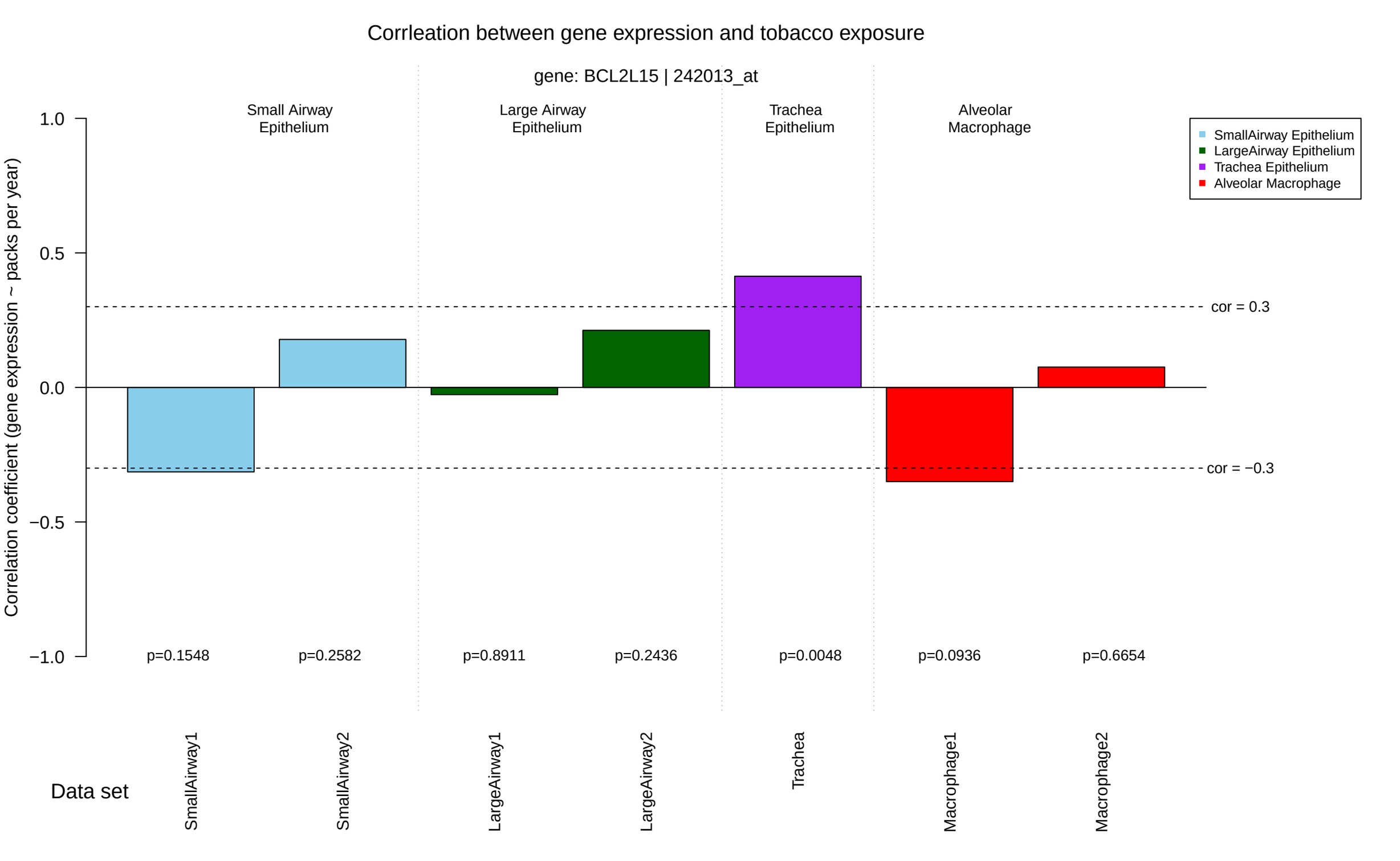

set: LargeAirway1 (p=0.4753)
BCL2L15 | 242013_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0658)
BCL2L15 | 242013_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0029)
BCL2L15 | 242013_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.005)
BCL2L15 | 242013_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4971)
BCL2L15 | 242013_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BCL2L15 | 242013_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1548
p=0.2582
p=0.8911
p=0.2436
p=0.0048
p=0.0936
p=0.6654
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2